Purpose

This article explains how DNA matches are identified and how shared autosomal DNA segments are used to estimate relationships between individuals.

Overview

Shared DNA segments are used to estimate relatedness by identifying genetic segments that individuals have in common. The more DNA two individuals share, the more closely they are related. DNA testers receive a list of matches showing the amount of shared DNA, measured in centimorgans (cM) and percentage.

Key Concepts

Define only what is necessary. Keep each definition tight.

- **Genome:** A genome is the complete set of DNA in an organism, containing the instructions needed for it to grow, function, and repair itself.
- **Chromosome:** A chromosome is a threadlike structure in the cell nucleus made of a long DNA molecule wrapped around proteins. It stores and organizes genetic information.
- **Single Nucleotide Polymorphism (SNP, pronounced “snip”):** A variation at a single position in DNA (A, T, C, or G). It is the most common type of genetic variation among people.
- **DNA Segment**
  DNA segments are continuous stretches of inherited genetic material, measured in base pairs, centimorgans (cM), or SNPs.
- **Centimorgan (cM)**
  A unit used to measure the length of DNA segments, displayed as a measurement of shared DNA between matches.
- **Total Shared DNA**
  The amount of DNA that individuals have in common, displayed in cMs and as a percentage.
- **Largest Common Segment**
  The largest DNA segment that two matching individuals have in common.

How DNA Matching Works

The MyHeritage DNA test analyses segments of your autosomal DNA - the first 22 of the 23 chromosomes. Chromosomes are inherited in pairs - one from each parent. Each chromosome consists of a sequence of four chemical bases: adenine (A), thymine (T), guanine (G), and cytosine (C), attached to a sugar called deoxyribose, and this forms a genetic code.

Within this sequence, small variations known as single nucleotide polymorphisms (SNPs) occur at specific positions on the genome. While most DNA is shared across individuals, these variations are used in genetic testing to identify differences between people. Autosomal DNA tests typically analyse hundreds of thousands of SNPs across the genome.

DNA segments are continuous stretches of these SNPs. When two individuals share the same segments, it indicates a common ancestor. This is referred to as having segments that are Identical By Descent (IBD).

A core feature of most autosomal DNA tests is the DNA Matches database, which displays other users with whom you share varying amounts of DNA.

Understanding Relationship Estimates

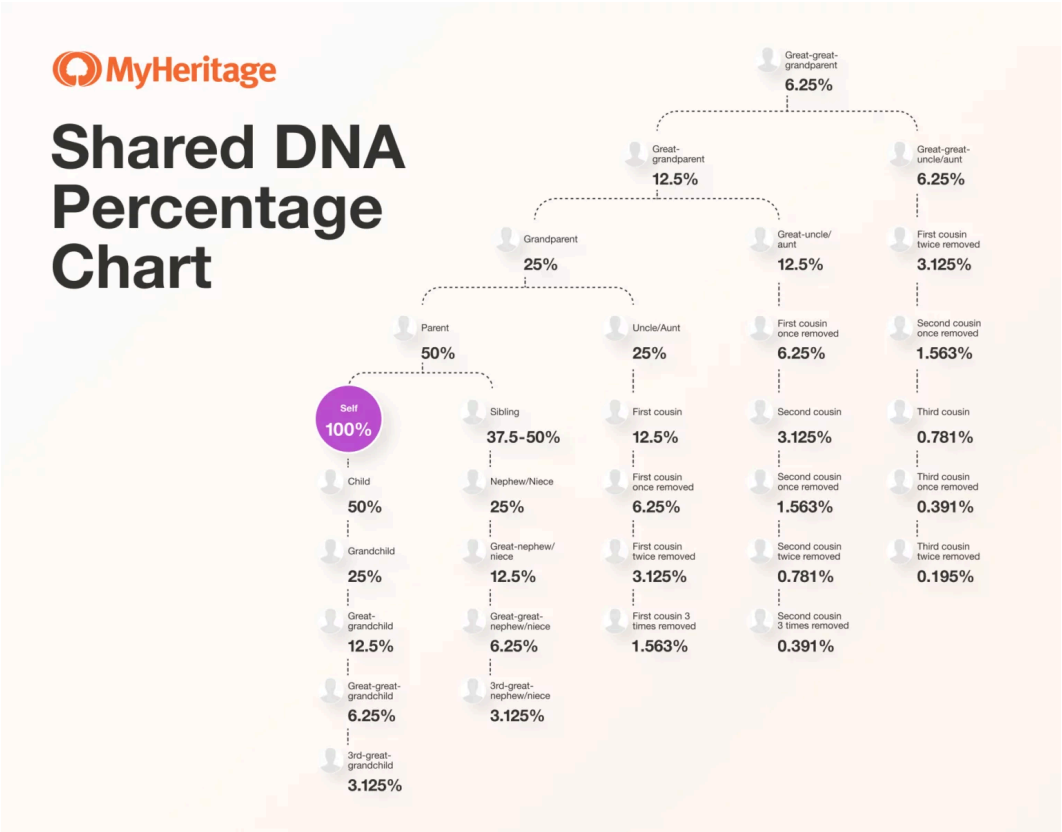
MyHeritage tests approximately 7044 centimorgans of your DNA segments in total. Individuals in the DNA Matches database share varying amounts of DNA with you, depending on how closely related you are.

Parents typically share around 50% of their DNA with you, although results may appear slightly lower (e.g. 48-49% or 3400-3500 cMs) due to limitations in testing and data interpretation, and the illegibility of a small number of segments. Grandparents, as well as aunts and uncles, will generally share about 25% with you.

Full siblings usually share approximately 30-40% of their DNA. This is because most autosomal DNA tests on the market do not fully test regions of DNA that are identical on both copies of a chromosome - as is the case with full siblings. However, identical (monozygotic) twins will show 99-100% shared DNA, as will a duplicate upload of the same DNA profile.

Half siblings, who share only one parent, typically share similar amounts of DNA to grandparents, aunts, and uncles. First cousins generally share around 12.5%.

As a general rule, the more distant the common ancestor, the less DNA is shared. Refer to the diagram below for typical ranges of shared DNA across different types of relatives:



What Affects Match Validity

To be considered a DNA match, two individuals must share at least 8 centimorgans (cM) of total DNA, with a largest shared segment of at least 6 cM.

Setting a minimum threshold for the largest segment helps distinguish true genetic relationships (usually up to fifth or sixth cousins) from coincidental similarities between people of the same population group. Without this threshold, individuals may appear to match due to many small shared segments that are identical by population (IBP), rather than inherited from a recent common ancestor.

IBP segments are more common in endogamous populations, such as Ashkenazi Jewish communities, Mennonites or Acadians, as an example. Therefore, the structure of shared DNA is also taken into account, and not just the total amount.

Matches based on many small segments are less reliable than those with fewer, longer segments. Very small segments are more likely to be identical by chance or population rather than inherited from a recent common ancestor. Certain regions of the genome are also more prone to shared DNA across unrelated individuals, which can result in false matches.

Common Questions

Keep answers short, controlled, and practical.

Why do I share different amounts of DNA with similar relatives?

Although siblings inherit half of their DNA from each parent, they do not inherit the exact same segments. As a result, siblings may share slightly different amounts of DNA with one another, typically within a similar range.

The same principle applies across generations. The children of siblings (first cousins) may also share varying amounts of DNA, falling within a defined range rather than a fixed value. This pattern continues with more distant relatives, such as second cousins, where the amount of shared DNA becomes smaller and more variable over time.

Why is my known relative showing as a more distant match?

The further away individuals are genetically related, the less DNA they share. Taking into account the relationship logic explained in the previous question, one can share varying amounts with relatives.

In some cases, this may be explained by a difference between the expected and biological relationships in a family tree. This is sometimes referred to as a “non-paternity event” (NPE), where a parent in the recorded pedigree is not the biological parent.

When this occurs, individuals who are assumed to be closely related may, in fact, share only one biological parent or a more distant connection. For example, siblings who share only one parent are half siblings and typically share around 20-25% of their DNA. Their children would be half-cousins, sharing approximately 6%, with even lower amounts shared in subsequent generations.

As a result, the amount of shared DNA may appear lower than expected based on the documented family relationship.

Can a DNA match be incorrect?

Yes. More distant matches - such as those in the fourth or fifth cousin range - may sometimes appear as “false positives.”

This can occur when a matching system identifies shared DNA segments that are common within a broader population group, rather than inherited from a recent common ancestor. These segments are referred to as *identical by population* (IBP), rather than *identical by descent* (IBD).

To reduce the likelihood of such matches, testing companies apply minimum thresholds for total shared DNA and segment size. These thresholds help prioritise more reliable matches and minimise the inclusion of coincidental similarities.

Why don’t some relatives appear as matches?

Autosomal DNA tests typically detect relatives up to about the fifth cousin level (approximately six generations). However, this is not a fixed range. DNA inheritance is random, and more distant relatives may not share enough detectable DNA to be identified as matches.

In some cases, differences between documented and biological relationships - sometimes referred to as non-paternity events (NPEs) - may also result in relatives appearing more distant than expected, or not appearing as matches at all.

It is also important to note that testing companies maintain separate databases. If a relative tested with a different company, they will not appear in your match list unless their DNA data has been uploaded to the same platform.

Limitations and Important Notes

- For many DNA test providers, genetic relationships are based on statistical analysis of shared DNA segments, rather than fixed values. As a result, relationship predictions are given as ranges rather than exact labels. Different relatives may share similar amounts of DNA, which can lead to overlapping relationship estimates. An example of this is the 20-25% range of shared DNA which includes: grandparents / grandchildren, aunts / uncles / nieces / nephews, and half siblings.
- Autosomal DNA tests can only detect relatives who share enough DNA to meet the platform’s matching thresholds. More distant relatives may not share detectable segments due to the random nature of inheritance. In addition, relatives will only appear as matches if they have tested with, or uploaded their DNA to, the same platform.

- DNA uploaded from a different company to the testing platform may provide different results due to various reasons: although companies test approximately 600-700,000 SNPs, they do not necessarily test the exact same SNPs. Also, imputation methods of DNA data differ between testing platforms.
- DNA results are most effective combined with traditional genealogical research methods. Historical records, family trees, and documented relationships help provide context for interpreting matches.

When to Investigate Further

Certain types of results in your DNA matches may warrant further investigation, as they may reveal hitherto unknown aspects of a tester's family history. This includes:

- Unexpected close matches
- Missing expected relatives
- Large discrepancies in known relationships